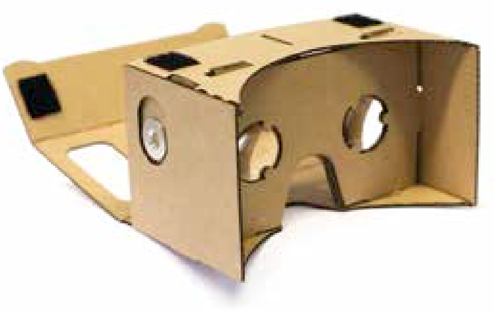
**Smart home, dumb stuff**
Meet Roost, a Wi-fi enabled 9V battery connected to a smoke detector.
**http://dgit.in/1LduTY3**

**Fun fact**
-40 degrees Farenheit is equal to -40 degrees Celsius
**http://dgit.in/1EdLZiF**



# Build an Oculus Rift on the cheap

## Here's what you can DO to experience real 3D gaming for just ₹1,000 or less!

By Anirudh Regidi
***anirudh.regidi@9dot9.in***

The Oculus Rift is currently the last word in gaming awesomeness but it's still six months from release and very expensive. Here's a temporary stand-in that won't break the bank!

The Oculus Rift was designed by a teenager in a garage who spent his time there tinkering with broken phones. If he can do it, why can't you? To save you the trouble of having to conduct years of research, let's first understand what an Oculus Rift does and how it does it. This knowledge will help us design our own version of the rift.

### What is a Rift?

**A head-tracker:** The Rift tracks your head movement with extreme precision. It does this using numerous infra-red LEDs that are affixed to the Rift in a precise pattern. A camera tracks these dots and uses algorithms to figure out where you're looking. Earlier models of the Rift used an accelerometer and magnetometer to do the same thing. Whichever the method used, the head-tracking information is translated to player head-movement on the PC.

**HMD:** It is a head-mounted display (HMD) that's transmitting images from the computer to a screen to a secondary display mounted a few inches from your eyes.

### Head-tracking

This is the easy part. Just pick out the December 2014 edition of SKOAR from your archives or from the DVD and follow the instructions for creating your own infra-red head-tracking unit. Before you fish out the article, here are the things you'll need:

- 3xInfra-red LEDs
- Appropriate resistors
- Webcam (preferably a PS3Eye cam)
- Soldering gun and solder
- Cardboard / sheet metal / plastic

### Head-Mounted Display (budget option)

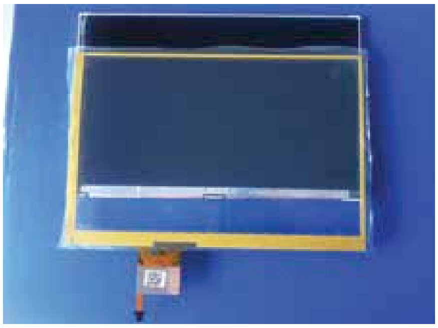
For the HMD, pick up a pair of cheap magnifying lenses and build a Google Cardboard device, or, pick up a clone at http://dgit.in/1FtggxZ for ₹400. Now you need to hook up the head-tracking clip and attach it to the Google Cardboard or its clone. You'll need to test it and calibrate it for a while, but once you're done, you can proceed to the next step.

The key to the Rift's ability to render 3D environments in "real" 3D is the driver. Unfortunately, not everyone has access to that driver and we don't have much of a choice but to go with free drivers that work only on a handful of games. Still, the experience of playing even those handful in 3D just can't be matched!

For a complete list of games that you can play in 3D with this setup, head over to http://dgit.in/1HA8mWV. This list includes some amazing games, including Crysis 3, Elite: Dangerous, Max Payne 3, DiRT Showdown and more.

There's a little bit of prep to be done, however, and it's as follows:

1. Install Splashtop Streamer from www.splashtop.com
2. Install the Splashtop 2.0 app on your smartphone.
3. Create a Splashtop account.
4. Set your games (from the list mentioned previously) to run in side-by-side (SBS) 3D mode. In this mode, two images are rendered, well, side-by-side.
5. Also set your game to run in windowed mode as Splashtop doesn't support full-screen apps as of yet.



If you're willing to dole out some cash, you can get yourself one of these, an actual 7" display

A more expensive, but hassle-free alternative to a TrackClip is the Arduino-based EDTracker

Google Cardboard is the perfect contraption for experiencing 3D on the cheap